EE2080 : Lecture 6 : 14 August 2025

PLAN: 1) Stopping Time Vs Non-stopping times : Examples.

2) Wald's Lemma.

3) Ergodicity

4) Gaussian Processes.

Examples of Stopping Time

1 Consider a random walk $\{X_n\}_{n\geq 0}$ with $X_0 = 0$ and $X_n = \sum_{i=1}^{n} Z_i$ where $Z_i \overset{iid}{\sim} Be(p)$

$$Z_i = \begin{cases} +1 & w/p \ p \\ -1 & w/p \ 1-p. \end{cases}$$

1) First time random walk reaches a level

$T = \min\{n \geq 1 : X_n = a\}$ ⟶ Stopping time

2) First return of a random walk to zero

$T = \min\{n \geq 1 : X_n = 0\}$ ⟶ Stopping time

3) Last time a random walk visits zero

$T = \max\{n \geq 1 : X_n = 0\}$ ⟶ Not a stopping time

4) One step before the random walk reaches a level.

$T_1 = \min\{n \geq 1 : X_n = a\}$

$T_2 \triangleq T_1 - 1$ ⟶ not a stopping time

(To check if $T_2 \leq n$, you have to visit the process till $n+1$)

5) Consider a Poisson arrival process in a cafe that opens at 8 am & closes at 8pm.

T = time at which half the day's customers have arrived

⟶ not a stopping time

Wald's Lemma

For a sequence of iid RVs $X_1, X_2, \dots$ and a stopping time N with finite expectations $\mathbb{E}[N] < \infty$ and $\mathbb{E}[X_1] < \infty$, the expected value of their random sum is

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[X_i]\,\mathbb{E}[N]$$

<u>Proof</u>: $S_N = \sum_{i=1}^{N} X_i$

We can rewrite this as the infinite sum $S_N = \sum_{i=1}^{\infty} X_i \mathbb{I}_{\{N \geq i\}}$

where $\mathbb{I}$ is the indicator RV

$$\therefore \mathbb{E}[S_N] = \mathbb{E}\left[\sum_{i=1}^{\infty} X_i \mathbb{I}_{\{N \geq i\}}\right] = \sum_{i=1}^{\infty} \mathbb{E}\left[X_i \mathbb{I}_{\{N \geq i\}}\right]$$

Since N is a stopping time, $\{N \geq i\}$ or the complementary event $\{N < i\}$ only depends on $X_1, \dots, X_{i-1}$

Since X_is are iid, $X_i \perp\!\!\!\perp \{X_1, \dots, X_{i-1}\}$ $\therefore X_i \perp\!\!\!\perp \{N \geq i\}$.

$$\Rightarrow \mathbb{E}\left[X_i \mathbb{I}_{\{N \geq i\}}\right] = \mathbb{E}[X_i]\,\underbrace{\mathbb{E}\left[\mathbb{I}_{\{N \geq i\}}\right]}_{P(N \geq i)}.$$

$$\therefore \mathbb{E}[S_N] = \sum_{i=1}^{\infty} \mathbb{E}[X_i]\,P(N \geq i) = \mathbb{E}[X_i] \sum_{i=1}^{\infty} P(N \geq i)$$

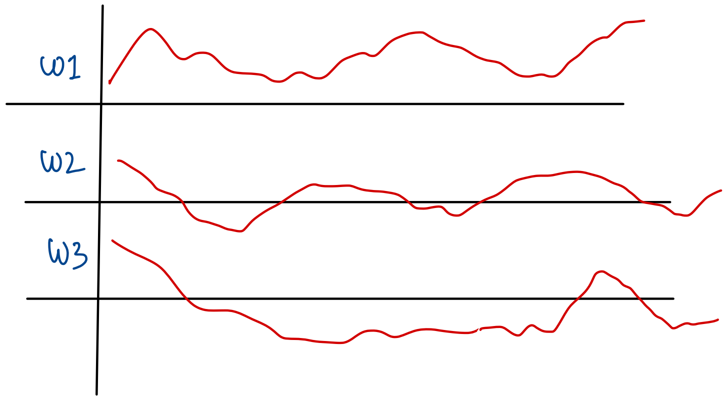
$$= \underline{\underline{\mathbb{E}[X_i]\,\mathbb{E}[N]}}$$

(N is positive-valued $\therefore$ By tail prob. formula of expectation)

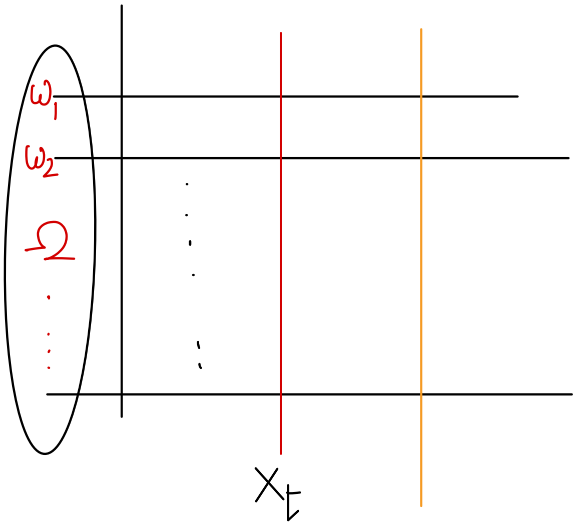
For a Gambler's ruin problem with initial wealth k & target wealth N_0; if prob. of winning at each step is $1/2$, Calculate the expected duration of the game. (Game ends when gambler attains target wealth or is ruined!).

— <u>Homework</u>

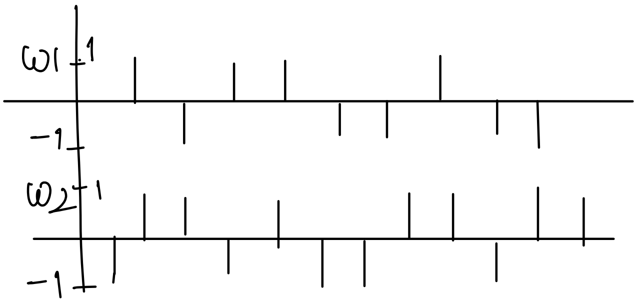
Ergodicity



$\{X_t, t \geq 0\}$ is a continuous valued continuous parameter process. Assume that it is stationary. From a single path, is it possible to deduce the statistical properties of the process?



What about this process $\{X_n, n \geq 0\}$



Ensemble average: $\mu_X(t) = \mathbb{E}[X_t] = \int x f_X(x)\,dx$. or $\sum x\, p_X(x)$.

Time Average (temporal): $\hat{\mu}_X(T) = \frac{1}{T}\int_{-T/2}^{T/2} X_\omega(t)\,dt \qquad \hat{\mu}_X = \lim_{T\to\infty} \hat{\mu}_X(T)$

Time averaged autocorrelation: $\hat{R}_{X_T}(\Delta) = \frac{1}{T}\int_{-T/2}^{T/2} X_\omega(t)\, X_\omega(t+\Delta)\,dt. \qquad \hat{R}_X(\Delta) = \lim_{T\to\infty} \hat{R}_{X_T}(\Delta)$

Ensemble averaged autocorrelation: $R_X(t,\Delta) = \mathbb{E}[X_t X_{t+\Delta}]$.

A stationary random process is said to be ergodic in mean if its time average converges to the ensemble average in the mean square sense: $\hat{\mu}_{X_T} \xrightarrow{m.s} \mathbb{E}[X]$. i.e $\hat{\mu}_X = \mathbb{E}[X]$

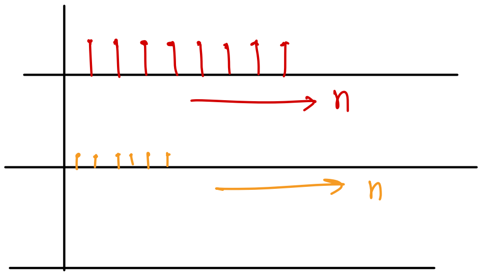
A stationary random process is said to be ergodic in autocorrelation if its time-averaged autocorrelation converges to the ensemble-averaged autocorrelation in the mean square sense.

$\hat{R}_{X_T}(\Delta) \xrightarrow{m.s} R_X(\Delta)$ i.e., $\hat{R}_X(\Delta) = R_X(\Delta)$.

Stationary but not ergodic: Example $(\Omega, \mathcal{F}, \mathbb{P})$

$\{X_n, n \in \mathbb{N}\}$ s.t For $\omega \in \Omega$, $X_n(\omega) = \omega \quad \forall n \in \mathbb{N}$

Is this process stationary? Is it ergodic?

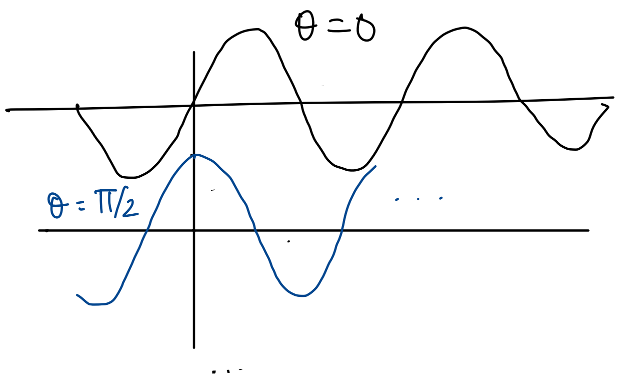


Example of an ergodic process

$X_t = A\sin(\omega_0 t + \Theta)$ $\quad \Theta \sim \text{unif}(0, 2\pi)$ $\quad A, \omega_0$ fixed positive constants

$$\hat{\mu}_{X_\theta}(T) = \frac{1}{T}\int_{-T/2}^{T/2} A\sin(\omega_0 t + \theta) \quad \underline{\underline{= 0}}$$

for some $\theta \in [0, 2\pi]$



$$E[X_t] = \int_0^{2\pi} A\sin(\omega_0 t + \theta)\frac{1}{2\pi}\,d\theta = 0$$

Properties of Autocorrelation:

Consider a W.S.S Process $\{X_t, t \in T\}$

1) $|R_{X,X}(\tau)| \leq R_{X,X}(0)$

$$\left|E[X_t X_{t+\tau}]\right| \leq \sqrt{E[X_t^2]\,E[X_{t+\tau}^2]}$$ Cauchy Schwarz

$$\sqrt{R_X(0)\,R_X(0)}$$

$$|R_X(\tau)| \leq R_X(0)$$

2) Assuming X_t is real, $R_X(\tau)$ is an even function of τ, i.e., $R_X(\tau) = R_X(-\tau)$

$$R_X(\tau) = E[X_t X_{t-\tau}]$$

$$R_X(-\tau) = E[X_{t-\tau} X_t]$$

Determine if the following processes are a) ergodic in mean
b) ergodic in autocorrelation

Homework

i) $X_n = A\cos(\omega_0 n + \Theta)$, $\Theta \sim \text{unif}(0, 2\pi)$, $n \in \mathbb{N} \cup \{0\}$.

A, ω_0 deterministic constants

ii) $X_n = A\cos(\omega_0 n + \Theta)$, $\Theta \sim \text{unif}(0, 2\pi)$, $n \in \mathbb{N} \cup \{0\}$.

A – a random variable independent of Θ

ω_0 – a deterministic constant

Gaussian Random Variables

– $X \sim \mathcal{N}(0,1)$ if $f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-x^2}{2}\right)$ $x \in (-\infty, \infty)$. Standard Normal

mean = 0, variance = 1

– $Y = \sigma X + \mu$, $Y \sim \mathcal{N}(\mu, \sigma^2)$ $f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$, $y \in (-\infty, \infty)$

Y - Gaussian RV with mean μ & variance σ^2.

– MGF of Y: $M_Y(s) = \mathbb{E}\left[\exp^{sY}\right] = \exp\left(s\mu + s^2\sigma^2/2\right)$

Gaussian Random Vectors : $\underline{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \sim \mathcal{N}(\underline{\mu_x}, K) \Rightarrow X_1, X_2, \ldots, X_n$ are jointly Gaussian R. Variables.

$$f_{\underline{X}}(\underline{x}) = \frac{1}{\sqrt{(2\pi)^N \det(K)}} \exp\left[\frac{-1}{2}(\underline{x} - \underline{\mu_x})^T K^{-1} (\underline{x} - \underline{\mu_x})\right] \quad \forall \underline{x} \in \mathbb{R}^n$$

K = Covariance matrix $\mathbb{E}\left[(\underline{X} - \underline{\mu_x})(\underline{X} - \underline{\mu_x})^T\right]$; $K_{i,j} = \mathbb{E}\left[(X_i - \mu_i)(X_j - \mu_j)\right]$

Gaussian Process : A stochastic process $\{X_t, t \in T\}$ is said to be a Gaussian process if for all $k \in \mathbb{N}$ and $t_1, t_2, \ldots, t_k \in T$, the set of random variables $X_{t_1}, X_{t_2}, \ldots, X_{t_k}$ is jointly Gaussian.

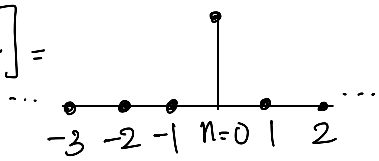
The covariance function: $C_X(s,t) = \mathbb{E}\left[(X_s - \mu_X(s))(X_t - \mu_X(t))\right]$, $\forall s,t \in T$

For a Gaussian process $\{X_t, t\in T\}$, the covariance function $C_X(s,t)$ and the mean $\mu_X(t)$ for all $s,t \in T$ completely specify the joint-pdf for all $(X_{t_1}, X_{t_2}, \dots, X_{t_k})$, $\forall k \in \mathbb{N}$, $t_1, t_2, \dots, t_k \in T$.

Example 1: Discrete-time IID Gaussian process

Discrete-time delta-fn.

$\delta[n] =$



$\{W_n, n \in \mathbb{Z}\}$, $W_n \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$

$$C_W(k,l) = \sigma^2 \delta[l-k]$$

$$f_{W_{n_1} \dots W_{n_k}}(w_1, \dots, w_k) = \frac{1}{(2\pi\sigma^2)^{k/2}} \exp\left(-\sum_{i=1}^{k} \frac{w_i}{2\sigma^2}\right)$$

Example 2: Discrete-time Gaussian sum-process.

$S_n = \sum_{i=1}^{n} W_i \longrightarrow$ linear combination of iid Gaussian RVs.

$\{S_n, n \geq 1\}$ - zero-mean Gaussian process.

For $k \leq l$, $C_S(k,l) = \mathbb{E}\left[\sum_{i=1}^{k} W_i \sum_{j=1}^{l} W_j\right] = \sum_{i=1}^{k} E[W_i^2] = k\sigma^2$

For any k,l $C_S(k,l) = \min(n,k)\,\sigma^2$.

Theorem: A Gaussian process $\{X_t, t \in T\}$ is stationary if and only if $\mu_X(t) = \mu_X(0)$ is a constant and $C_X(s,t) = C_X(0, t-s)$ $\forall t,s \in T$

$\Rightarrow$ A W.S.S Gaussian process is stationary!

The Wiener process / Brownian Motion (Example of a Gaussian Process).

Consider a stochastic process $\{X_t, t \in T\}$ where $T = \mathbb{Z}^+$ or $T = \mathbb{R}^+$.

i) $\{X_t, t \in T\}$ has stationary increments if for any $t_1 < t_2$, $X(t_2) - X(t_1) \stackrel{d}{=} X(t_2 - t_1) - X(0)$

ii) $\{X_t, t \in T\}$ has independent increments if for any $t_1 < t_2 < t_3 < \dots < t_k$, the RVs $X_{t_2} - X_{t_1}, X_{t_3} - X_{t_2}, \dots \; X_{t_k} - X_{t_{k-1}}$ are statistically independent

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Defn: A Wiener Process / Brownian motion is a zero-mean Gaussian process $\{X_t, t \geq 0\}$ which has the following properties

i) Stationary and independent increments

ii) $X_0 = 0$ with probability 1.

iii) has continuous sample functions with prob. 1, i.e X_t is almost surely continuous in t.